Answer **one** question from this section in the spaces provided.

8 **(a)** The kinetic theory of gases showed that the pressure p of an ideal gas is related to the mean-square speed $\langle c^2 \rangle$ of the gas molecules according to equation 8.1.

$$pV = \frac{1}{3}Nm\ \langle c^2 \rangle \qquad \text{(equation 8.1)}$$

In deriving equation 8.1, the force F exerted by N molecules, each of mass m, on one wall of a cubical container of volume V is given by equation 8.2.

$$F = \left(\frac{N}{t}\right)\Delta p \qquad \text{(equation 8.2)}$$

where Δp is the change in momentum of one gas molecule after colliding with the wall and $\left(\frac{N}{t}\right)$ is the rate of collision of the molecules with the same wall.

Two of the assumptions used in deriving equation 8.1 are:

Assumption 1: The duration of collision between each molecule and the wall is negligible compared to the time spent in between collision with the same wall.

Assumption 2: Total volume of the gas molecules is negligible compared to the volume occupied by the gas.

Each gas molecule is modelled as a hard sphere.

Explain how the rate of collision and the pressure exerted by the gas will defer from the values predicted by the kinetic theory if

(i) assumption 1 becomes invalid while the rest of the assumptions remained valid,

collision rate: .....

.....

..... [2]

gas pressure: .....

.....

..... [2]

(ii) assumption 2 becomes invalid while the rest of the assumptions remained valid.

collision rate: .....

.....

..... [2]

gas pressure: .....

.....

..... [2]

(b) A heat engine is a machine that uses the heat supplied to do mechanical work. The working substance is a fixed mass of ideal gas enclosed in a cylinder with a frictionless movable piston. The gas is made to undergo the changes in pressure p and volume V shown in Fig. 8.1.

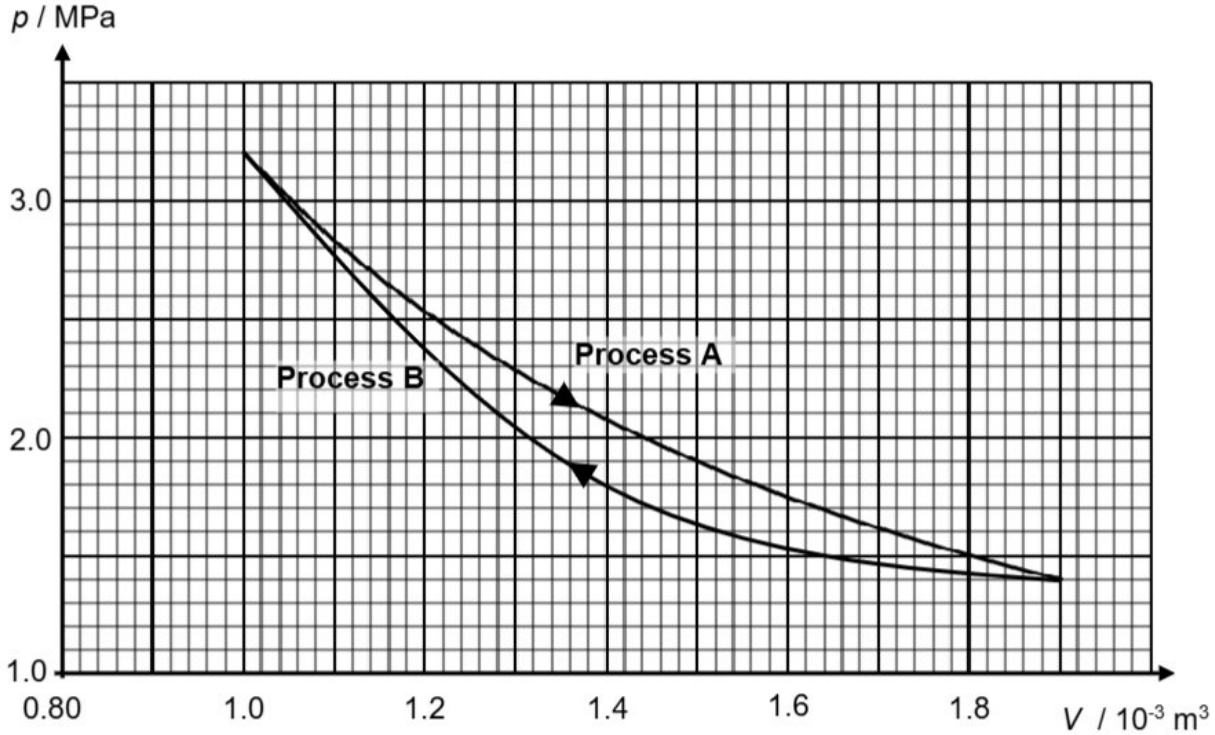


Fig. 8.1

(i) Use data from Fig. 8.1 to show quantitatively that process A cannot take place at constant temperature.

[2]

(ii) There is no transfer of thermal energy between the gas and the surrounding during process A.

Explain, using the kinetic theory of gases, why the temperature of the gas decreases during process A.

.....

.....

.....

.....

.....

..... [3]

(iii) The pressure and volume of the gas in process A are related according to equation 8.3.

$$pV^{\beta} = \text{constant} \qquad \text{(equation 8.3)}$$

where β is a constant.

1. Use data from Fig. 8.1 to determine the value of the constant β.

β = ... [2]

2. State the unit of β.

unit of β: ..... [1]

(iv) Estimate the work done on the gas during process B.

work done on gas = ... J [2]

(v) The efficiency of a heat engine is given by

$$\text{efficiency} = \frac{net\ work\ done\ by\ gas}{heat\ supplied} \times 100\%$$

Explain why a heat engine based on the cyclic process shown in Fig. 8.1 cannot exist in practice.

.....

.....